

Adobe Inc ADBE ★★★★★ 13 Jan 2026 22:29, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
327.65 USD 12 Jan 2026	560.00 USD 13 Jun 2025 02:52, UTC	0.59	129.74 USD Bil 13 Jan 2026	Wide	Large Value	High	Exemplary	 7 Jan 2026 06:00, UTC

Sector	Industry
 Technology	Software - Application

Business Description

Adobe provides content creation, document management, and digital marketing and advertising software and services to creative professionals and marketers for creating, managing, delivering, measuring, optimizing, and engaging with compelling content multiple operating systems, devices, and media. The company operates with three segments: digital media content creation, digital experience for marketing solutions, and publishing for legacy products (less than 5% of revenue).

have to be 400 basis points lower throughout our entire forecast, which seems highly unlikely.

► We continue to see a path for annual margin expansion coupled with solid but decelerating growth over the next five years, even if that can be uneven at times.

Coming up: Fiscal 2026 guidance was mixed relative to our expectations, with revenue slightly ahead and non-GAAP operating margin slightly light, while the first-quarter outlook was ahead on both measures.

► The one obvious blemish was that ARR grew nicely in the quarter and revenue guidance skewed positive, but the ARR outlook for fiscal 2026 calls for deceleration to 10.2% growth. We struggle to reconcile otherwise positive data points here.

Business Strategy & Outlook Dan Romanoff, CPA, Senior Equity Analyst, 11 Dec 2025

Adobe has come to dominate in content creation software with its iconic Photoshop and Illustrator solutions, both now part of the broader Creative Cloud. Over the years, the firm has added new products and features to the suite through organic development and bolt-on acquisitions to drive the most comprehensive portfolio of tools used in print, digital, and video content creation. The December 2021 launch of Adobe Express helps further broaden the company's funnel, as it incorporates popular features of the full Creative Cloud but comes in lower-cost and free versions. The 2023 introduction of Firefly marks an important artificial intelligence solution that should also attract new users and help extend the competitive position of the platform. We think Adobe is properly focusing on bringing new users under its umbrella and believe that converting these users will become more important over time.

CEO Shantanu Narayan provided Adobe with another growth leg in 2009 with the acquisition of Omniture, a leading web analytics solution that serves as the foundation of the digital experience segment that Adobe has used as a platform to layer in a variety of other marketing and advertising solutions. Adobe benefits from the natural cross-selling opportunity from Creative Cloud to the business and operational aspects of marketing and advertising.

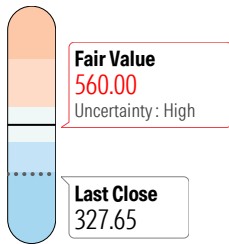
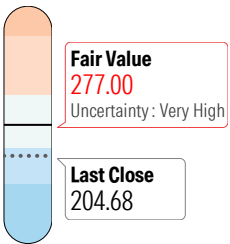
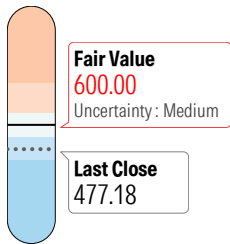
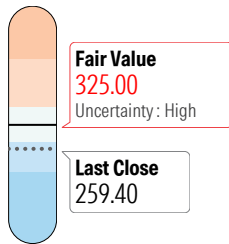
The Document Cloud is driven by one of Adobe's first products, Acrobat, and the ubiquitous PDF file format created by the company, and is now a multibillion-dollar business. The rise of smartphones and tablets, coupled with bring-your-own-device and a mobile workforce have made a file format that is usable on any screen more relevant than ever.

Adobe believes it is attacking an addressable market well in excess of \$200 billion. The company is introducing and leveraging features across its various cloud offerings to drive a more cohesive experience, win new clients, upsell users to higher price point solutions, and cross sell digital media offerings. We expect M&A efforts will continue to bolster all aspects of Adobe's portfolio, which it surely will do to defend against emerging competitors.

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Competitors

	Adobe Inc ADBE	Oracle Corp ORCL	Microsoft Corp MSFT	Salesforce Inc CRM
	 Fair Value 560.00 Uncertainty: High Last Close 327.65	 Fair Value 277.00 Uncertainty: Very High Last Close 204.68	 Fair Value 600.00 Uncertainty: Medium Last Close 477.18	 Fair Value 325.00 Uncertainty: High Last Close 259.40
Economic Moat	Wide	Wide	Wide	Wide
Currency	USD	USD	USD	USD
Fair Value	560.00 13 Jun 2025 02:52, UTC	277.00 18 Dec 2025 01:24, UTC	600.00 31 Jul 2025 03:51, UTC	325.00 29 May 2025 01:49, UTC
1-Star Price	868.00	484.75	810.00	503.75
5-Star Price	336.00	138.50	420.00	195.00
Assessment	Undervalued 13 Jan 2026	Undervalued 13 Jan 2026	Undervalued 13 Jan 2026	Undervalued 13 Jan 2026
Morningstar Rating	★★★★★ 13 Jan 2026 22:29, UTC	★★★★★ 13 Jan 2026 22:28, UTC	★★★★★ 13 Jan 2026 22:28, UTC	★★★★★ 13 Jan 2026 22:32, UTC
Analyst	Dan Romanoff, Senior Equity Analyst	Luke Yang, Equity Analyst	Dan Romanoff, Senior Equity Analyst	Dan Romanoff, Senior Equity Analyst
Capital Allocation	Exemplary	Standard	Exemplary	Standard
Price/Fair Value	0.59	0.73	0.78	0.74
Price/Sales	5.89	9.71	12.12	6.20
Price/Book	11.80	19.63	9.77	4.07
Price/Earning	15.65	41.28	34.00	33.52
Dividend Yield	0.00%	0.98%	0.71%	0.64%
Market Cap	129.74 Bil	581.21 Bil	3,498.20 Bil	225.87 Bil
52-Week Range	307.91—465.70	118.86—345.72	344.79—555.45	221.96—367.09
Investment Style	Large Value	Large Growth	Large Blend	Large Blend

Bulls Say Dan Romanoff, CPA, Senior Equity Analyst, 11 Dec 2025

- Adobe is the de facto standard in content creation software and PDF file editing, categories it created and still dominates.
- Adobe continues to introduce new solutions and features that defend its moat and drive innovation. Express has helped develop a funnel of new users while Firefly is a capable new visual content generation tool.
- Adobe is extending its empire in the creative world from content creation to marketing services more broadly through the expansion of its digital experience segment. This segment should drive growth in the coming years.

Bears Say Dan Romanoff, CPA, Senior Equity Analyst, 11 Dec 2025

- Competition has increased in recent years with the rise of both Canva and Figma, so Adobe's

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competition position is weakened.

- Digital experience is an emerging space and one that Adobe neither created nor dominates. Growth could be slower than we anticipate or margin expansion may not materialize.
- Growth has slowed meaningfully with the rise of both new competitors and generative AI. There is a fear that AI models can replace applications like Adobe's or pressure the seat licensing model.

Economic Moat Dan Romanoff, CPA, Senior Equity Analyst, 14 Jun 2024

For Adobe overall, we assign a wide economic moat rating arising from switching costs. Based on the company's segments, we believe digital media has a wide moat based on switching costs and digital experience has a narrow moat also arising from switching costs. We believe Adobe's moat will allow the company to earn returns in excess of its cost of capital over the next 20 years.

Switching costs for software are often driven by several factors, in our view. The more critical the function and the more touch points across an organization a software vendor has, the higher the switching costs. There is also the direct time and expense of implementing a new software package for the customer while maintaining the existing platform and retraining employees on a new system. Additionally, there is operational risk of changing software vendors, including business process re-engineering, loss of data during the changeover, and overall project execution. A major implementation is likely to involve a system integrator and can take in excess of a year in bad cases. Lastly, lost productivity is likely to be an issue as customers move up a learning curve on the new system along with the distraction of users involved in the function where the change is occurring.

Retention metrics typically help inform investors on both the presence and the durability of a moat. These come in two flavors: gross, which describes what percentage of the customer base remains customers after a given period, and net, which highlights what percentage of customer spending is retained by the software provider after a given period. Adobe does not provide retention metrics because its user base has a large percentage of small-business users and periodic subscribers. Software firms regularly see lower retention rates for small-business users than for enterprise customers. Further, in the case of Adobe, many creative professionals work on a contract basis, meaning that they might subscribe to Creative Cloud for a month, not subscribe for the next month, and subscribe again the following month. We believe Adobe experiences customer retention of 85%-90%, but we also do not think this is particularly relevant for the company, given the dynamics at play with the customer base.

We believe the digital media segment enjoys a wide moat based on switching costs. Digital media represents approximately 70%-75% of revenue. This segment contains Creative Cloud, which is nearly 50% of total revenue, and Document Cloud, which is approximately 10% of revenue. Creative Cloud is composed of iconic products Photoshop, Premiere, Illustrator, and After Effects, among others, and a variety of mobile versions of these products and additional discrete mobile solutions. Document Cloud

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consists of the Acrobat family of products, including Sign.

We believe Creative Cloud enjoys the widest moat under the Adobe umbrella. Like most software solutions, once Creative Cloud or any of the individual applications contained therein is inserted into the workflow of creative professionals, it becomes very difficult to change solutions. Further, while there is no shortage of competitive point solutions, Adobe Creative Cloud is so pervasive in the creative world that replacing it would be an insurmountable barrier, in our view. Because nearly all creative professionals rely on the Creative Cloud, all other creative professionals must also use it. While the Creative Cloud has its issues, particularly premium pricing, and any one organization or freelance professional might be willing to switch, they would find it difficult to work in an industry that has generally standardized around it. Similarly, Adobe's large installed base helps ensure that when the company releases or acquires a related new solution, it immediately faces a substantial and welcoming audience.

Since its introduction in the late 1980s, Photoshop quickly became the industry leader and eventually the industry standard for image editing software. Rather than remaining complacent, Adobe has consistently invested in the solution, introducing new features and adding applications that could be sold to existing users of Photoshop. These features and products were both internally developed and from acquisitions. For these reasons, Creative Cloud still finds itself at the forefront of the industry.

We believe Adobe's creation of the PDF file format, its first-mover advantage with Acrobat, and significant installed base have created a wide moat based on switching costs for the firm's Document Cloud. Within Document Cloud, Adobe created the portable document format as an evolution of its original product, PostScript. In 1990, there was no file format that was readily usable across operating system platforms, but by the mid-1990s, there were several products vying for widespread adoption. Adobe's PDF won, thanks in part to distributing Acrobat Reader for free to personal computer original equipment manufacturers, and has become the standard. Adobe then developed an enterprise Acrobat product as a PDF editor and workflow solution. We believe there are no truly competitive solutions to the PDF file format, even if there is a wide variety of PDF editors in free and paid versions. By any measure, Acrobat remains the gold standard in PDF editors. Over the years, the firm has added a variety of key features to Acrobat, including e-signatures, which became mainstream overnight during the covid lockdowns.

In our opinion, the digital experience segment enjoys a narrow moat based on switching costs. Digital experience represents approximately 25% of revenue. This segment contains Adobe Experience Platform, Adobe Analytics, Adobe Experience Manager, Marketo Engage, and Adobe Commerce, among other solutions. We view Digital Experience as a natural extension of the digital media segment and see the same switching costs that we see for most software companies. Once these applications are integrated into a business process workflow, we believe they are difficult to change. A digital

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experience platform, by its very nature, is complex, involving numerous applications and touching a wide range of systems and data repositories used by the customer.

Unlike the creative side, Adobe does not have a first-mover advantage in digital experience. The company made its foray into this broad umbrella initially through acquisitions, notably of Omniture in 2009. Since then, it has made other meaningful acquisitions, such as Marketo and Magento, both in 2018, and has also organically added solutions to the portfolio. Through these efforts, Adobe has established itself as a leader in various categories that fall under the digital experience umbrella, including digital marketing analytics, campaign management, and customer engagement, among others.

Digital experience solutions are relatively new compared with the creative side, and unlike with Photoshop, where no serious competitive threats exist, there are a wide variety of large competitors in the various marketing analytics, campaign management, customer engagement, advertising platform, and related areas. While this has been a nascent and rapidly emerging area, more recently we see several companies starting to emerge as leaders with a holistic and complete platform. One of these leaders is Adobe. We believe Adobe's tightly integrated and robust platform, with its accelerated innovation based on customer feedback from its extensive creative customer base, is seen as a strong suite by the marketplace and well positioned to capture an outsize position in this young and growing market.

Fair Value and Profit Drivers Dan Romanoff, CPA, Senior Equity Analyst, 11 Dec 2025

Our fair value estimate for Adobe is \$560 per share, which implies a fiscal 2026 enterprise value/sales multiple of 9 times, and an adjusted P/E multiple of 24 times.

We model a five-year revenue CAGR of approximately 9%. We model modestly decelerating growth in both digital media and digital experience over time. Digital experience should benefit both from price increases that should filter in over the course of several years and increasing penetration into an enormous market as defined by Adobe. We believe a relatively frictionless cross-selling opportunity exists for the company, as creative professionals are already steeped in Adobe products. The desire to consolidate vendors makes Adobe an obvious choice to turn to for needed marketing software solutions, and the fact that Adobe's products are strong should help initially in what we believe is a large greenfield opportunity. Within digital media, we have been impressed by Adobe's ability to draw in new users and infuse the platform with generative AI features. Additionally, the company has had success upselling existing users to higher price point products and cross selling acquired technologies, such as Workfront and Frame.io. We believe continued innovation, gathering new users, and upselling existing users in Creative Cloud should help drive strong growth for the next several years.

We model non-GAAP operating margins, which were 46% in fiscal 2025, to expand modestly on an

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annual basis given increased scale. Management continues to talk about pushing margins higher over time but given the company's margins are already right near the top of the software group, we think expansion is likely but will be more incremental from here.

Risk and Uncertainty Dan Romanoff, CPA, Senior Equity Analyst, 11 Dec 2025

We assign Adobe a Morningstar Uncertainty Rating of High. It faces risks that vary by segment. Creative Cloud's dominant market share over the last 25 years means a significant portion of high-margin revenue is at risk, however small that risk may be, if a competitor were to make inroads in the space. Figma and Canva are the two most prominent competitors, and both are relatively young compared with Adobe. The dampening of cross-selling opportunities with digital experience would likely then be diminished, which would be problematic, as digital experience represents the larger growth opportunity over the next five years in our view. While Adobe is generally considered a leader in the various categories included under its digital experience umbrella, it did not create any of these categories and does not dominate them the way it does with Creative Cloud.

Adobe has built the digital experience business through acquisitions. The two recent acquisitions of Magento and Marketo also pose risks, as those were on the larger side for the company. Any integration missteps could delay new contract signings. Further, material missteps could possibly result in substantial write-downs regarding these (or other) acquisitions. Further, while the margin structure may ultimately be lower in digital experience relative to Creative Cloud, the company has worked to improve margins over time, and we believe Adobe must continue to drive down costs and expand margins to meet investor expectations.

Lastly, there is a general fear that AI will either obviate the need for Adobe's software altogether or at least pressure the seat count in the firm's subscription. While this fear spans most software subscription models, the fear seems to be especially acute for Adobe.

While we do not see significant ESG risks, we note that Adobe faces strong competition for software engineers on the hiring front and also faces risks arising from a potential data breach within its data centers.

Capital Allocation Dan Romanoff, CPA, Senior Equity Analyst, 11 Dec 2025

We assign Adobe an Exemplary Capital Allocation Rating. This reflects our assessments of sound balance sheet, exceptional investment, and appropriate shareholder distributions. We think investments back into the business are most likely to be the key driver of total shareholder returns and are therefore appropriately prioritized over other capital returns such as dividends and buyback, although given the company's prodigious free cash flow generation, we see share buybacks and acquisitions continuing.

Adobe's capital deployment strategy centers around re-investing in the business and making generally small tuck-in acquisitions. We expect the balance sheet to remain sound as the company has typically

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maintained a conservative financial position and generates more than enough free cash flow to fund growth, buy back shares, and execute its acquisition strategy.

Shantanu Narayen joined the company in 1998 as senior vice president of worldwide product development. He was promoted to COO, and ultimately to CEO, a position he has had since 2007. Narayen has steered the company through a variety of critical changes over the years, including the Creative Suite bundle, the transition to a SaaS model, and the Omniture acquisition that created the Digital Experience segment in 2009.

Analyst Notes Archive

Adobe: Impressive AI Innovation on Display at Max User Conference Dan Romanoff, CPA, Senior Equity Analyst, 29 Oct 2025

Adobe hosted its annual Max user conference highlighting recent product innovations, and hosted a brief investor session where the company reaffirmed previously issued fourth-quarter guidance. Why it matters: Product announcements and demonstrations were impressive, including Firefly Model 5, Firefly Foundry, Project Moonlight (an artificial intelligence assistant), Graph (a new work flow tool), and the full integration of third-party models directly within the platform. Adobe also announced strategic partnerships with Google. Nearly all products were immediately available or are in beta testing, with Moonlight coming soon. Management also noted free unlimited image generation for certain higher-end users through Dec. 1, which we think should generate significant interest, especially given the new tools. The bottom line: We maintain our fair value estimate of \$560 per share for wide-moat Adobe and view shares as attractively valued. We do not think the firm has erased investor concerns, but we see momentum building under the surface in terms of both product innovation and sales execution. After good third-quarter results in September and an upbeat and impressive Max user conference, we sense Adobe has turned the corner. The stock peaked in January 2024 at approximately \$635 and has been in a slide since, seemingly bottoming around the \$350 area since July 2025. Slowing sales growth had fanned investor fears around the threat generative AI posed to seat counts and the overall need for Adobe's tools in general. Adobe emphatically rejected the notion that there was seat count pressure, pointing to burgeoning growth in various products, including Firefly AI. Coming up: Management reiterated fourth-quarter guidance and said the fourth quarter was progressing well. The firm plans to begin changing the performance and guidance metrics it provides in 2026, with both new and old measures to be provided during the transition throughout the upcoming year.

Adobe Earnings: The Sky Is Still Not Falling as AI and New Products Gain Momentum Dan Romanoff, CPA, Senior Equity Analyst, 12 Sep 2025

Adobe's third-quarter fiscal 2025 revenue grew by 10% year over year in constant currency to \$5.99 billion, while non-GAAP operating margin was 46.3%, both of which topped guidance. Fourth-quarter

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guidance was slightly ahead of expectations on the top and bottom lines. Why it matters: This marks the fifth consecutive quarter of revenue upside relative to our expectations and seems to be a turning point for the company where results, artificial intelligence traction, and guidance are all pointing in the same positive direction. We see momentum building for Adobe's AI solutions. Twenty-nine billion images have been created now, demonstrating good adoption. Adobe has already easily surpassed its \$250 million goal for AI annual recurring revenue, which is a full quarter ahead of management's target. The bottom line: We maintain our fair value estimate of \$560 for wide-moat Adobe and view shares as attractively valued. Our estimates through fiscal 2026 are near consensus, so we do not view our assumptions as exceptional. The two main controversies around the stock center on emerging competition and threats from AI. To support the current stock price, both risks would need to drive revenue growth lower by 400 basis points annually throughout our entire forecast, which we view as an extreme scenario. We continue to see a path for annual margin expansion coupled with solid but decelerating growth over the next five years, even if that might be uneven at times. Coming up: Fourth-quarter guidance is in line with our expectations, which marks the third straight quarter of solid guidance. The company will provide guidance for fiscal 2026 on its fourth-quarter call, which will be more important than the fourth-quarter results. The firm will hold its annual Adobe MAX event in late October, including an investor session on Oct. 28. We expect the event to focus on AI and other product-related announcements, but we are not expecting much in the way of financial updates.

Adobe Earnings: Good Quarter but Overt Signs of AI Strength Are Elusive to Investors Dan Romanoff, CPA, Senior Equity Analyst, 13 Jun 2025

Adobe's second-quarter revenue grew by 11% year over year in constant currency to \$5.87 billion, while non-GAAP operating margin was 45.5%, both of which topped guidance. Adobe also raised its full-year guidance on both the top and bottom lines. Why it matters: This marks the fourth consecutive quarter of revenue upside relative to our expectations, as digital media was strong and drove the vast majority of revenue upside. Adobe continues to push tiered pricing throughout the suite, with Firefly now having tiers from free up to \$200 per user per month. Management is bullish on artificial intelligence usage, which is up to 24 billion images created now and believes underlying trends are accelerating. We estimate Adobe has in excess of \$160 million in annual recurring revenue from AI, while the firm noted it remains on pace to top \$250 million in AI annual recurring revenue exiting the year. The bottom line: We lower our fair value estimate to \$560 per share, from \$590 previously, as we lowered our growth assumptions by approximately 30 basis points annually over the next five years. Shares are attractive in our view. We think the controversy around the stock revolves around AI supplanting Adobe's solutions. In order to support the current stock price we would need to lower our revenue growth by 300 basis points annually throughout our entire forecast, which we do not think is reasonable. We continue to see a path for annual margin expansion coupled with solid but decelerating growth over the next five years. Coming up: We view the raised annual guidance positively, as management tends to be conservative.

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That said, the \$135 million increase at the midpoint is the same as the cumulative upside relative to the midpoint of guidance for the first half of the year. Third-quarter guidance was as expected. We note that the original 2025 outlook included a \$200 million currency headwind, which has since eased but there has been no associated increase in guidance.

Adobe: The Relentless Focus on Generative AI Is Producing Innovation and Spurring Adoption Dan Romanoff, CPA, Senior Equity Analyst, 19 Mar 2025

Wide-moat Adobe's upbeat investor session focused mainly on artificial intelligence innovation and monetization, along with its new customer-focused strategy. Management stopped short of declaring concrete long-term financial targets, but did provide some details consistent with our model for the next five years. Thus, we hold our model steady and keep our fair value estimate of \$590 per share. Shares remain attractive, in our view. We continue to view Adobe as the clear leader in software solutions for the creative industry and believe the company is well-positioned to capitalize on the ongoing shift toward generative AI. Adobe Summit 2025 reinforced this thesis, in our view. From a financial perspective, management said it aspires to increase revenue by double digits annually while expanding EPS at a faster rate. Our model is a hair under this revenue aspiration, but we think it's consistent with the spirit of the firm's goals. We see growth driven by new customer additions, new pricing models for generative AI—tiered pricing specifically, and a drumbeat of new solution releases such as the day's Orchestrator announcement. One interesting data point was that about 1,500 of its 22,000 enterprise customers pay for five or more of Adobe's products, so there is tremendous room for cross-selling efforts. On the margin side, we model approximately 200 basis points of non-GAAP operating margin expansion over the next five years. Our margin assumption is based on declining research and development intensity as generative AI matures. Management added that share buybacks should shrink the share count in the coming years, which also contributes to EPS growth that outpaces revenue growth. From an AI standpoint, Adobe continues to integrate Firefly and generative AI throughout the portfolio, introduce new AI solutions, and evolve on pricing models for these solutions. We continue to believe generative AI will be a tailwind for growth and supportive of the firm's moat.

Adobe Earnings: Quietly Stacking Good Quarters As AI Momentum Builds Dan Romanoff, CPA, Senior Equity Analyst, 13 Mar 2025

We maintain our fair value estimate of \$590 per share after wide-moat Adobe reported good first-quarter results, provided second-quarter guidance that was in line with our expectations, and reaffirmed its previously issued outlook for fiscal 2025. We left our model largely unchanged after this quarterly update. We are encouraged by recent product introductions throughout the portfolio and think Firefly leaves Adobe well-positioned in artificial intelligence. We also see monetization in the early stages. Along those lines, Adobe disclosed that AI is generating \$125 million in annual recurring revenue, or ARR, which it expects to double by the end of the year. We continue to think recent price

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increases, product launches, and rapid generative AI adoption should help drive growth in 2025. We observe a growing disconnect between our fair value estimate and current trading levels that makes little sense with stable fundamentals and the bear case of AI obviating the need for Creative Cloud not materializing once again. With three straight quarters of revenue upside, we see the demand environment as stable and see signs that the company's AI strategy is working. Adobe's strategy of creating a broad demand funnel at the top with Express and driving AI usage to convert users seems to be resonating. The firm's new products are gaining traction, notably with Acrobat AI Assistant, Firefly, and GenStudio, which make the \$125 million AI book of business. First-quarter revenue grew 10% year over year as reported and 11% in constant currency to \$5.71 billion, exceeding the top end of guidance at \$5.68 billion. Management's ability to drive margins in the face of high investment levels surrounding generative AI remains impressive. We think margins can grind a little higher over time but will be limited by Adobe's already stellar profitability. Non-GAAP operating margin was 47.5%, compared with 47.6% a year ago.

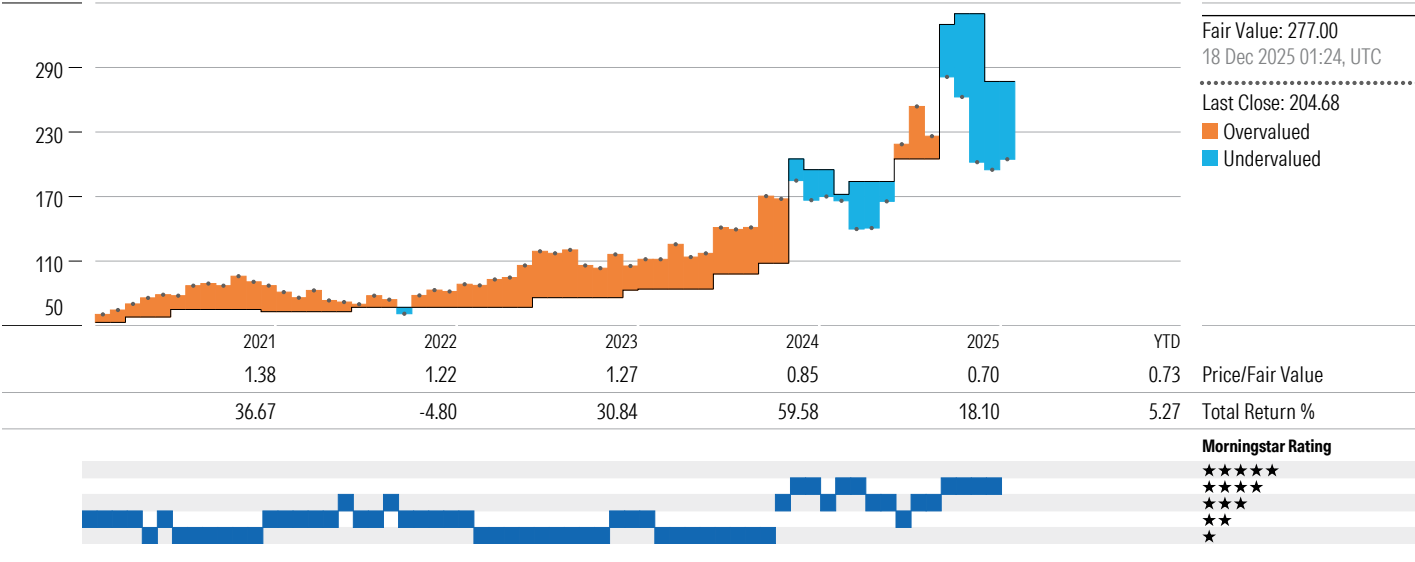
Adobe Earnings: A Good Quarter With Mixed Signals, but Document Cloud Shines Dan Romanoff, CPA, Senior Equity Analyst, 12 Dec 2024

We are lowering our fair estimate to \$590 per share, from \$635, after wide-moat Adobe reported good fourth-quarter results and an outlook for fiscal 2025 that we believe is mildly disappointing. Based on guidance, we lowered our 2025 estimates, which also drove a smoothing over several years. While we do not see a meaningful change to our long-term thinking, we see some nicks in the firm's armor. Given another selloff, we observe a clear disconnect between management's excitement and the internal signs of success that they see relative to what investors are seeing. We are encouraged by recent product introductions throughout the portfolio and think Firefly leaves Adobe well-positioned in artificial intelligence. We are also encouraged by the firm's decision to use pricing as a lever within a coming tiered pricing structure tied largely to AI for core applications. We think recent price increases, product launches, and rapid generative AI adoption should help drive growth in 2025. Considering our fair value cut and the stock selloff, we still see shares as undervalued. The demand environment appears relatively healthy between the upside last quarter and this quarter, and we see Adobe leaning into innovation. Adobe's strategy of creating a broad demand funnel at the top with Express and driving AI usage to convert users seems to be resonating. The company's new products are gaining traction, notably with Acrobat AI Assistant, Firefly, and GenStudio. Management noted good performance in all products through all channels. Fourth-quarter revenue increased 11% year over year in constant currency and as reported to \$5.61 billion, exceeding the top end of guidance at \$5.55 billion. Management is pushing investors to focus on the entire book of business rather than just net new digital media annual recurring revenue, which were up 14% and 2% year over year, respectively, and diverge in their support for management's enthusiasm. ■■■

Adobe Inc ADBE ★★★★★ 13 Jan 2026 22:29, UTC

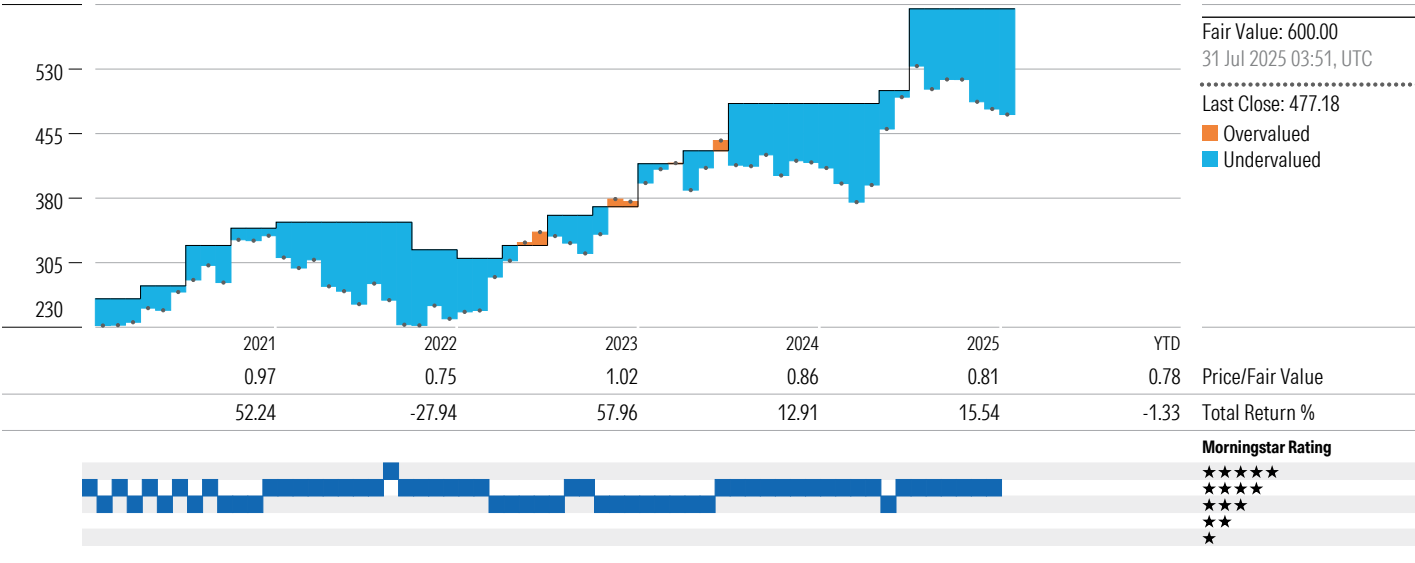
Competitors Price vs. Fair Value

Oracle Corp ORCL



Total Return % as of 12 Jan 2026. Last Close as of 12 Jan 2026. Fair Value as of 18 Dec 2025 01:24, UTC.

Microsoft Corp MSFT



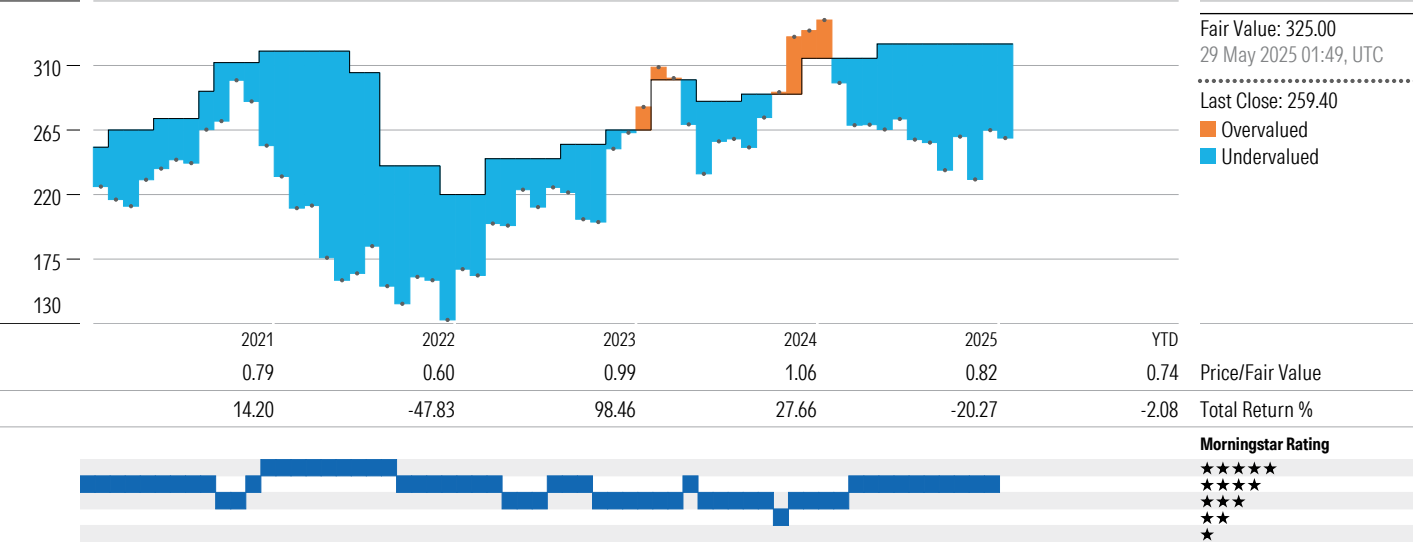
Total Return % as of 12 Jan 2026. Last Close as of 12 Jan 2026. Fair Value as of 31 Jul 2025 03:51, UTC.

Adobe IncADBE★★★★★

13 Jan 2026 22:29, UTC

Competitors Price vs. Fair Value

Salesforce IncCRM



Total Return % as of 12 Jan 2026. Last Close as of 12 Jan 2026. Fair Value as of 29 May 2025 01:49, UTC.

Adobe Inc ADBE ★★★★★ 13 Jan 2026 22:29, UTC

Last Price 327.65 USD 12 Jan 2026	Fair Value Estimate 560.00 USD 13 Jun 2025 02:52, UTC	Price/FVE 0.59	Market Cap 129.74 USD Bil 13 Jan 2026	Economic Moat™ Wide	Equity Style Box Large Value	Uncertainty High	Capital Allocation Exemplary	ESG Risk Rating Assessment¹ 7 Jan 2026 06:00, UTC
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Morningstar Valuation Model Summary

Financials as of 10 Dec 2025

Fiscal Year, ends 30 Nov	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Revenue (USD Mil)	19,409	21,505	23,769	26,000	28,312	30,821	33,539	36,471
Operating Income (USD Mil)	6,650	6,741	8,706	9,329	10,427	11,512	12,795	14,242
EBITDA (USD Mil)	7,522	7,598	9,524	10,109	11,135	12,097	13,399	14,920
Adjusted EBITDA (USD Mil)	7,522	7,598	9,524	10,109	11,135	12,097	13,399	14,920
Net Income (USD Mil)	5,428	5,560	7,130	7,366	8,365	9,364	10,437	11,619
Adjusted Net Income (USD Mil)	7,377	8,281	8,941	9,541	10,663	11,992	13,160	14,430
Free Cash Flow To The Firm (USD Mil)	6,780	9,787	8,354	7,794	6,618	9,130	10,142	11,341
Weighted Average Diluted Shares Outstanding (Mil)	459	450	427	403	395	395	395	395
Earnings Per Share (Diluted) (USD)	11.82	12.36	16.70	18.28	21.18	23.71	26.42	29.41
Adjusted Earnings Per Share (Diluted) (USD)	16.07	18.42	20.94	23.67	27.00	30.36	33.32	36.53
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Margins & Returns as of 10 Dec 2025

	3 Year Avg	Actual			Forecast					5 Year Avg
		2023	2024	2025	2026	2027	2028	2029	2030	
Operating Margin %	34.1	34.3	31.4	36.6	35.9	36.8	37.4	38.2	39.1	37.5
EBITDA Margin %	38.1	38.8	35.3	40.1	38.9	39.3	39.3	40.0	40.9	39.7
Adjusted EBITDA Margin %	—	38.8	35.3	40.1	38.9	39.3	39.3	40.0	40.9	39.7
Net Margin %	28.0	28.0	25.9	30.0	28.3	29.5	30.4	31.1	31.9	30.2
Adjusted Net Margin %	38.0	38.0	38.5	37.6	36.7	37.7	38.9	39.2	39.6	38.4
Free Cash Flow To The Firm Margin %	38.5	34.9	45.5	35.2	30.0	23.4	29.6	30.2	31.1	28.9

Growth & Ratios as of 10 Dec 2025

	3 Year CAGR	Actual			Forecast					5 Year CAGR
		2023	2024	2025	2026	2027	2028	2029	2030	
Revenue Growth %	10.5	10.2	10.8	10.5	9.4	8.9	8.9	8.8	8.7	8.9
Operating Income Growth %	12.6	9.0	1.4	29.2	7.2	11.8	10.4	11.2	11.3	10.3
EBITDA Growth %	4.9	8.2	1.0	25.4	6.1	10.2	8.6	10.8	11.4	12.2
Adjusted EBITDA Growth %	11.1	8.2	1.0	25.4	6.1	10.2	8.6	10.8	11.4	9.4
Earnings Per Share Growth %	18.3	17.1	4.6	35.1	9.4	15.9	12.0	11.5	11.3	12.0
Adjusted Earnings Per Share Growth %	18.3	17.2	14.6	13.7	13.0	14.0	12.5	9.7	—	12.0

Valuation as of 10 Dec 2025

	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Price/Earning	32.1	17.4	14.8	13.1	11.5	10.2	9.3	—
Price/Sales	14.3	10.6	5.5	5.0	4.6	4.2	3.9	—
Price/Book	14.3	10.2	11.4	9.8	6.6	4.4	3.2	—
Price/Cash Flow	—	—	—	—	—	—	—	—
EV/EBITDA	36.5	29.8	13.6	12.8	11.7	10.7	9.7	—
EV/EBIT	41.3	33.5	149.1	13.9	12.4	11.3	10.1	—
Dividend Yield %	—	—	—	—	—	—	—	—
Dividend Payout %	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Free Cash Flow Yield %	—	—	—	—	—	—	—	—

Operating Performance / Profitability as of 10 Dec 2025

Fiscal Year, ends 30 Nov	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
ROA %	17.5	18.2	18.4	24.2	23.3	22.6	19.7	17.8
ROE %	33.9	32.9	39.4	61.3	58.1	45.4	33.7	27.3
ROIC %	25.5	27.0	27.8	35.3	36.2	37.9	39.0	41.5

Adobe Inc ADBE ★★★★★ 13 Jan 2026 22:29, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
327.65 USD 12 Jan 2026	560.00 USD 13 Jan 2025 02:52, UTC	0.59	129.74 USD Bil 13 Jan 2026	Wide	Large Value	High	Exemplary	 7 Jan 2026 06:00, UTC

Financial Leverage (Reporting Currency)

Fiscal Year, ends 30 Nov	Actual			Forecast				
	2023	2024	2025	2026	2027	2028	2029	2030
Debt/Capital %	1.5	2.6	—	2.5	1.8	1.7	1.3	0.8
Assets/Equity	1.8	2.1	2.5	2.5	2.0	1.7	1.5	1.4
Net Debt/EBITDA	-0.5	-0.2	—	0.0	-0.5	-1.3	-1.9	-2.5
Total Debt/EBITDA	0.5	0.8	—	0.6	0.4	0.4	0.3	0.2
EBITDA/ Net Interest Expense	-50.5	-40.0	-340.1	168.3	-393.3	130.8	199.8	204.6

Forecast Revisions as of 10 Dec 2025

Prior data as of 12 Sep 2025	2026		2027		2028	
	Current	Prior	Current	Prior	Current	Prior
Fair Value Estimate Change (Trading Currency)	560.00	560.09	—	—	—	—
Revenue (USD Mil)	26,000	25,776	28,312	28,149	30,821	30,722
Operating Income (USD Mil)	9,329	9,333	10,427	10,387	11,512	11,582
EBITDA (USD Mil)	10,109	10,106	11,135	11,091	12,097	12,197
Net Income (USD Mil)	9,541	9,767	10,663	10,918	11,992	12,017
Earnings Per Share (Diluted) (USD)	18.28	18.05	21.18	20.21	23.71	22.60
Adjusted Earnings Per Share (Diluted) (USD)	23.67	23.37	27.00	26.12	30.36	28.75
Dividends Per Share (USD)	0.00	0.00	0.00	0.00	0.00	0.00

Key Valuation Drivers as of 10 Dec 2025

Cost of Equity %	9.0
Pre-Tax Cost of Debt %	5.3
Weighted Average Cost of Capital %	8.8
Long-Run Tax Rate %	18.0
Stage II EBI Growth Rate %	8.8
Stage II Investment Rate %	30.0
Perpetuity Year	20

Additional estimates and scenarios available for download at <https://pitchbook.com/>.

Discounted Cash Flow Valuation as of 10 Dec 2025

	USD Mil
Present Value Stage I	67,386
Present Value Stage II	57,067
Present Value Stage III	101,487
Total Firm Value	225,940
Cash and Equivalents	5,431
Debt	6,210
Other Adjustments	0
Equity Value	225,161
Projected Diluted Shares	403
Fair Value per Share (USD)	560.00

Adobe Inc ADBE ★★★★★ 13 Jan 2026 22:29, UTC

Last Price	Fair Value Estimate	Price/FVE	Market Cap	Economic Moat™	Equity Style Box	Uncertainty	Capital Allocation	ESG Risk Rating Assessment¹
327.65 USD 12 Jan 2026	560.00 USD 13 Jun 2025 02:52, UTC	0.59	129.74 USD Bil 13 Jan 2026	Wide	Large Value	High	Exemplary	 7 Jan 2026 06:00, UTC

ESG Risk Rating Breakdown

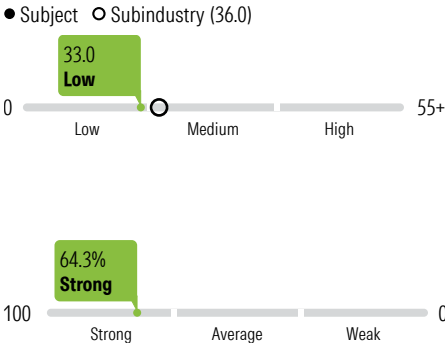
Exposure

Company Exposure¹	33.0
- Manageable Risk	30.7
Unmanageable Risk²	2.3

Management

Manageable Risk	30.7
- Managed Risk³	19.7
Management Gap⁴	11.0

Overall Unmanaged Risk 13.3



- ▶ Exposure represents a company's vulnerability to ESG risks driven by their business model
- ▶ Exposure is assessed at the Subindustry level and then specified at the company level
- ▶ Scoring ranges from 0-55+ with categories of low, medium, and high-risk exposure
- ▶ Management measures a company's ability to manage ESG risks through its commitments and actions
- ▶ Management assesses a company's efficiency on ESG programs, practices, and policies
- ▶ Management score ranges from 0-100% showing how much manageable risk a company is managing

ESG Risk Rating



ESG Risk Ratings measure the degree to which a company's value is impacted by environmental, social, and governance risks, by evaluating the company's ability to manage the ESG risks it faces.

1. A company's Exposure to material ESG issues 2. Unmanageable Risk refers to risks that are inherent to a particular business model that cannot be managed by programs or initiatives 3. Managed Risk = Manageable Risk multiplied by a Management score of 64.3% 4. Management Gap assesses risks that are not managed, but are considered manageable 5. ESG Risk Rating Assessment = Overall Unmanaged Risk = Management Gap plus Unmanageable Risk

ESG Risk Rating Assessment⁵



ESG Risk Rating is of Jan 07, 2026. Highest Controversy Level is as of Jan 08, 2026. Sustainalytics Subindustry: Enterprise and Infrastructure Software. Sustainalytics provides Morningstar with company ESG ratings and metrics on a monthly basis and as such, the ratings in Morningstar may not necessarily reflect current Sustainalytics' scores for the company. For the most up to date rating and more information, please visit: sustainalytics.com/esg-ratings/.

Peer Analysis 07 Jan 2026

Company Name	Exposure	Management	ESG Risk Rating
Adobe Inc	33.0 Low 0 —●— 55+	64.3 Strong 100 —●— 0	13.3 Low 0 —●— 40+
Microsoft Corp	32.8 Low 0 —●— 55+	60.3 Strong 100 —●— 0	14.4 Low 0 —●— 40+
Salesforce Inc	34.2 Low 0 —●— 55+	56.1 Strong 100 —●— 0	16.3 Low 0 —●— 40+
Oracle Corp	35.0 Low 0 —●— 55+	60.0 Strong 100 —●— 0	15.4 Low 0 —●— 40+
DoubleVerify Holdings Inc	40.1 Medium 0 —●— 55+	45.8 Average 100 —●— 0	23.1 Medium 0 —●— 40+

Appendix

Historical Morningstar Rating

Adobe Inc ADBE 13 Jan 2026 22:29, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	—	—	—	—	★★★★★
Dec 2025 ★★★★	Nov 2025 ★★★★★	Oct 2025 ★★★★	Sep 2025 ★★★★	Aug 2025 ★★★★	Jul 2025 ★★★★	Jun 2025 ★★★★	May 2025 ★★★★	Apr 2025 ★★★★	Mar 2025 ★★★★	Feb 2025 ★★★★	Jan 2025 ★★★★
Dec 2024 ★★★★	Nov 2024 ★★★★	Oct 2024 ★★★★	Sep 2024 ★★★★	Aug 2024 ★★★	Jul 2024 ★★★	Jun 2024 ★★★★	May 2024 ★★★★	Apr 2024 ★★★★	Mar 2024 ★★★★	Feb 2024 ★★★	Jan 2024 ★★★
Dec 2023 ★★★	Nov 2023 ★★	Oct 2023 ★★★	Sep 2023 ★★★	Aug 2023 ★★★	Jul 2023 ★★★	Jun 2023 ★★★	May 2023 ★★★	Apr 2023 ★★★	Mar 2023 ★★★	Feb 2023 ★★★★	Jan 2023 ★★★★
Dec 2022 ★★★★	Nov 2022 ★★★★	Oct 2022 ★★★★	Sep 2022 ★★★★★	Aug 2022 ★★★★	Jul 2022 ★★★★	Jun 2022 ★★★★	May 2022 ★★★★★	Apr 2022 ★★★★★	Mar 2022 ★★★★	Feb 2022 ★★★★	Jan 2022 ★★★★
Dec 2021 ★★★★	Nov 2021 ★★★	Oct 2021 ★★★	Sep 2021 ★★★	Aug 2021 ★★	Jul 2021 ★★★	Jun 2021 ★★★	May 2021 ★★★	Apr 2021 ★★★	Mar 2021 ★★★★	Feb 2021 ★★★	Jan 2021 ★★★

Oracle Corp ORCL 13 Jan 2026 22:28, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	—	—	—	—	★★★★
Dec 2025 ★★★★	Nov 2025 ★★★★	Oct 2025 ★★★★	Sep 2025 ★★★	Aug 2025 ★★★	Jul 2025 ★★	Jun 2025 ★★★	May 2025 ★★★	Apr 2025 ★★★★	Mar 2025 ★★★★	Feb 2025 ★★★	Jan 2025 ★★★★
Dec 2024 ★★★★	Nov 2024 ★★★	Oct 2024 ★	Sep 2024 ★	Aug 2024 ★	Jul 2024 ★	Jun 2024 ★	May 2024 ★	Apr 2024 ★	Mar 2024 ★	Feb 2024 ★★	Jan 2024 ★★
Dec 2023 ★★	Nov 2023 ★	Oct 2023 ★	Sep 2023 ★	Aug 2023 ★	Jul 2023 ★	Jun 2023 ★	May 2023 ★	Apr 2023 ★	Mar 2023 ★	Feb 2023 ★★	Jan 2023 ★★
Dec 2022 ★★	Nov 2022 ★★	Oct 2022 ★★	Sep 2022 ★★★★	Aug 2022 ★★	Jul 2022 ★★	Jun 2022 ★★★★	May 2022 ★★	Apr 2022 ★★	Mar 2022 ★★	Feb 2022 ★★	Jan 2022 ★★
Dec 2021 ★	Nov 2021 ★	Oct 2021 ★	Sep 2021 ★	Aug 2021 ★	Jul 2021 ★	Jun 2021 ★★	May 2021 ★	Apr 2021 ★★	Mar 2021 ★★	Feb 2021 ★★	Jan 2021 ★★

Microsoft Corp MSFT 13 Jan 2026 22:28, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	—	—	—	—	★★★★
Dec 2025 ★★★★	Nov 2025 ★★★★	Oct 2025 ★★★★	Sep 2025 ★★★★	Aug 2025 ★★★★	Jul 2025 ★★★★	Jun 2025 ★★★	May 2025 ★★★★	Apr 2025 ★★★★	Mar 2025 ★★★★	Feb 2025 ★★★★	Jan 2025 ★★★★
Dec 2024 ★★★★	Nov 2024 ★★★★	Oct 2024 ★★★★	Sep 2024 ★★★★	Aug 2024 ★★★★	Jul 2024 ★★★★	Jun 2024 ★★★	May 2024 ★★★	Apr 2024 ★★★★	Mar 2024 ★★★	Feb 2024 ★★★★	Jan 2024 ★★★
Dec 2023 ★★★	Nov 2023 ★★★	Oct 2023 ★★★★	Sep 2023 ★★★★	Aug 2023 ★★★	Jul 2023 ★★★★	Jun 2023 ★★★	May 2023 ★★★	Apr 2023 ★★★★	Mar 2023 ★★★★	Feb 2023 ★★★★	Jan 2023 ★★★★
Dec 2022 ★★★★	Nov 2022 ★★★★	Oct 2022 ★★★★	Sep 2022 ★★★★★	Aug 2022 ★★★★	Jul 2022 ★★★★	Jun 2022 ★★★★	May 2022 ★★★★	Apr 2022 ★★★★	Mar 2022 ★★★★	Feb 2022 ★★★★	Jan 2022 ★★★★
Dec 2021 ★★★	Nov 2021 ★★★	Oct 2021 ★★★	Sep 2021 ★★★★	Aug 2021 ★★★	Jul 2021 ★★★★	Jun 2021 ★★★	May 2021 ★★★★	Apr 2021 ★★★★	Mar 2021 ★★★★	Feb 2021 ★★★★	Jan 2021 ★★★★

Salesforce Inc CRM 13 Jan 2026 22:32, UTC

Dec 2026	Nov 2026	Oct 2026	Sep 2026	Aug 2026	Jul 2026	Jun 2026	May 2026	Apr 2026	Mar 2026	Feb 2026	Jan 2026
—	—	—	—	—	—	—	—	—	—	—	★★★★
Dec 2025	Nov 2025	Oct 2025	Sep 2025	Aug 2025	Jul 2025	Jun 2025	May 2025	Apr 2025	Mar 2025	Feb 2025	Jan 2025
★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★
Dec 2024	Nov 2024	Oct 2024	Sep 2024	Aug 2024	Jul 2024	Jun 2024	May 2024	Apr 2024	Mar 2024	Feb 2024	Jan 2024
★★★	★★	★★★	★★★	★★★	★★★	★★★	★★★★	★★★	★★★	★★★	★★★
Dec 2023	Nov 2023	Oct 2023	Sep 2023	Aug 2023	Jul 2023	Jun 2023	May 2023	Apr 2023	Mar 2023	Feb 2023	Jan 2023
★★★	★★★	★★★★	★★★★	★★★★	★★★	★★★	★★★	★★★★	★★★★	★★★★	★★★★
Dec 2022	Nov 2022	Oct 2022	Sep 2022	Aug 2022	Jul 2022	Jun 2022	May 2022	Apr 2022	Mar 2022	Feb 2022	Jan 2022
★★★★	★★★★	★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★	★★★★★
Dec 2021	Nov 2021	Oct 2021	Sep 2021	Aug 2021	Jul 2021	Jun 2021	May 2021	Apr 2021	Mar 2021	Feb 2021	Jan 2021
★★★★	★★★	★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★	★★★★

Research Methodology for Valuing Companies

Overview

At the heart of our valuation system is a detailed projection of a company's future cash flows, resulting from our analysts' research. Analysts create custom industry and company assumptions to feed income statement, balance sheet, and capital investment assumptions into our globally standardized, proprietary discounted cash flow, or DCF, modeling templates. We use scenario analysis, in-depth competitive advantage analysis, and a variety of other analytical tools to augment this process. Moreover, we think analyzing valuation through discounted cash flows presents a better lens for viewing cyclical companies, high-growth firms, businesses with finite lives (e.g., mines), or companies expected to generate negative earnings over the next few years. That said, we don't dismiss multiples altogether but rather use them as supporting cross-checks for our DCF-based fair value estimates. We also acknowledge that DCF models offer their own challenges (including a potential proliferation of estimated inputs and the possibility that the method may miss short-term market-price movements), but we believe these negatives are mitigated by deep analysis and our long-term approach.

Morningstar's equity research group ("we," "our") believes that a company's intrinsic worth results from the future cash flows it can generate. The Morningstar Rating for stocks identifies stocks trading at a discount or premium to their intrinsic worth—or fair value estimate, in Morningstar terminology. Five-star stocks sell for the biggest risk adjusted discount to their fair values, whereas 1-star stocks trade at premiums to their intrinsic worth.

Four key components drive the Morningstar rating: (1) our assessment of the firm's economic moat, (2) our estimate of the stock's fair value, (3) our uncertainty around that fair value estimate and (4) the current market price. This process ultimately culminates in our single-point star rating.

1. Economic Moat

The concept of an economic moat plays a vital role not only in our qualitative assessment of a firm's long-term investment potential, but also in the actual calculation of our fair value estimates. An economic moat is a structural feature that allows a firm to sustain excess profits over a

long period of time. We define economic profits as returns on invested capital (or ROIC) over and above our estimate of a firm's cost of capital, or weighted average cost of capital (or WACC). Without a moat, profits are more susceptible to competition. We have identified five sources of economic moats: intangible assets, switching costs, network effect, cost advantage, and efficient scale.

Companies with a narrow moat are those we believe are more likely than not to achieve normalized excess returns for at least the next 10 years. Wide-moat companies are those in which we have very high confidence that excess returns will remain for 10 years, with excess returns more likely than not to remain for at least 20 years. The longer a firm generates economic profits, the higher its intrinsic value. We believe low-quality, no-moat companies will see their normalized returns gravitate toward the firm's cost of capital more quickly than companies with moats.

When considering a company's moat, we also assess whether there is a substantial threat of value destruction, stemming from risks related to ESG, industry disruption, financial health, or other idiosyncratic issues. In this context, a risk is considered potentially value destructive if its occurrence would eliminate a firm's economic profit on a cumulative or midcycle basis. If we deem the probability of occurrence sufficiently high, we would not characterize the company as possessing an economic moat.

2. Estimated Fair Value

Combining our analysts' financial forecasts with the firm's economic moat helps us assess how long returns on invested capital are likely to exceed the firm's cost of capital. Returns of firms with a wide economic moat rating are assumed to fade to the perpetuity period over a longer period of time than the returns of narrow-moat firms, and both will fade slower than no-moat firms, increasing our estimate of their intrinsic value.

Our model is divided into three distinct stages:

Stage I: Explicit Forecast

In this stage, which can last five to 10 years, analysts make full financial statement forecasts, including items such as revenue, profit margins, tax rates, changes in working capital accounts, and capital spending. Based on these projections, we calculate earnings before interest,

after taxes (EBI) and the net new investment (NNI) to derive our annual free cash flow forecast.

Stage II: Fade

The second stage of our model is the period it will take the company's return on new invested capital—the return on capital of the next dollar invested ("RONIC")—to decline (or rise) to its cost of capital. During the Stage II period, we use a formula to approximate cash flows in lieu of explicitly modeling the income statement, balance sheet, and cash flow statement as we do in Stage I. The length of the second stage depends on the strength of the company's economic moat. We forecast this period to last anywhere from one year (for companies with no economic moat) to 10–15 years or more (for wide-moat companies). During this period, cash flows are forecast using four assumptions: an average growth rate for EBI over the period, a normalized investment rate, average return on new invested capital (RONIC), and the number of years until perpetuity, when excess returns cease. The investment rate and return on new invested capital decline until a perpetuity value is calculated. In the case of firms that do not earn their cost of capital, we assume marginal ROICs rise to the firm's cost of capital (usually attributable to less reinvestment), and we may truncate the second stage.

Stage III: Perpetuity

Once a company's marginal ROIC hits its cost of capital, we calculate a continuing value, using a standard perpetuity formula. At perpetuity, we assume that any growth or decline or investment in the business neither creates nor destroys value and that any new investment provides a return in line with estimated WACC.

Because a dollar earned today is worth more than a dollar earned tomorrow, we discount our projections of cash flows in stages I, II, and III to arrive at a total present value of expected future cash flows. Because we are modeling free cash flow to the firm—representing cash available to provide a return to all capital providers—we discount future cash flows using the WACC, which is a weighted average of the costs of equity, debt, and preferred stock (and any other funding sources), using expected future proportionate long-term, market-value weights.

3. Uncertainty Around That Fair Value Estimate

Morningstar's Uncertainty Rating is designed to capture the range of potential outcomes for a company's intrinsic value. This rating is used to assign the margin of safety required before investing, which in turn explicitly drives our stock star rating system. The Uncertainty Rating is aimed at identifying the confidence we should have in assigning a fair value estimate for a given stock.

Our Uncertainty Rating is meant to take into account anything that can increase the potential dispersion of future

Morningstar Equity Research Star Rating Methodology



Research Methodology for Valuing Companies

outcomes for the intrinsic value of a company, and anything that can affect our ability to accurately predict these outcomes. The rating begins with a suggested rating produced by a quantitative process based on the trailing 12-month standard deviation of daily stock returns. An analyst overlay is then applied, with analysts using the suggested rating, historical rating data, and their own knowledge of the company to inform them as they make the final Uncertainty Rating decision. Ultimately, the rating decision rests with the analyst. Analysts take into account many characteristics when making their final decision, including cyclical factors, operational and financial factors such as leverage, company-specific events, ESG risks, and anything else that might increase the potential dispersion of future outcomes and our ability to estimate those outcomes.

Our recommended margin of safety—the discount to fair value demanded before we'd recommend buying or selling the stock—widens as our uncertainty of the estimated value of the equity increases. The more uncertain we are about the potential dispersion of outcomes, the greater the discount we require relative to our estimate of the value of the firm before we would recommend the purchase of the shares. In addition, the Uncertainty Rating provides guidance in portfolio construction based on risk tolerance.

Our Uncertainty Ratings are: Low, Medium, High, Very High, and Extreme.

Margin of Safety		
Qualitative Analysis	★★★★★ Rating	★ Rating
Uncertainty Ratings		
Low	20% Discount	25% Premium
Medium	30% Discount	35% Premium
High	40% Discount	55% Premium
Very High	50% Discount	75% Premium
Extreme	75% Discount	300% Premium

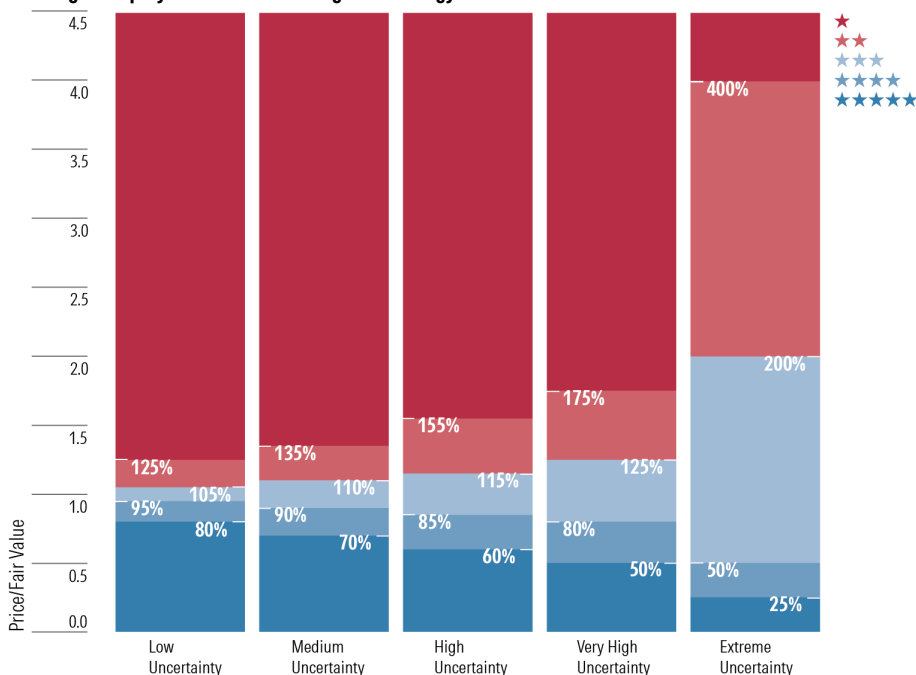
Our uncertainty rating is based on the interquartile range, or the middle 50% of potential outcomes, covering the 25th percentile–75th percentile. This means that when a stock hits 5 stars, we expect there is a 75% chance that the intrinsic value of that stock lies above the current market price. Similarly, when a stock hits 1 star, we expect there is a 75% chance that the intrinsic value of that stock lies below the current market price.

4. Market Price

The market prices used in this analysis and noted in the report come from exchange on which the stock is listed which we believe is a reliable source.

For more details about our methodology, please go to <https://shareholders.morningstar.com>

Morningstar Equity Research Star Rating Methodology



Morningstar Star Rating for Stocks

Once we determine the fair value estimate of a stock, we compare it with the stock's current market price on a daily basis, and the star rating is automatically re-calculated at the market close on every day the market on which the stock is listed is open. Our analysts keep close tabs on the companies they follow, and, based on thorough and ongoing analysis, raise or lower their fair value estimates as warranted.

Please note, there is no predefined distribution of stars. That is, the percentage of stocks that earn 5 stars can fluctuate daily, so the star ratings, in the aggregate, can serve as a gauge of the broader market's valuation. When there are many 5-star stocks, the stock market as a whole is more undervalued, in our opinion, than when very few companies garner our highest rating.

We expect that if our base-case assumptions are true the market price will converge on our fair value estimate over time generally within three years (although it is impossible to predict the exact time frame in which market prices may adjust).

Our star ratings are guideposts to a broad audience and individuals must consider their own specific investment goals, risk tolerance, tax situation, time horizon, income needs, and complete investment portfolio, among other factors.

The Morningstar Star Ratings for stocks are defined below:

★★★★★ We believe appreciation beyond a fair risk-adjusted return is highly likely over a multiyear time frame. Scenario analysis developed by our analysts indicates that the current market price represents an excessively pessimistic outlook, limiting downside risk and maximizing upside potential.

★★★★ We believe appreciation beyond a fair risk-adjusted return is likely.

★★★ Indicates our belief that investors are likely to receive a fair risk-adjusted return (approximately cost of equity).

★★ We believe investors are likely to receive a less than fair risk-adjusted return.

★ Indicates a high probability of undesirable risk-adjusted returns from the current market price over a multi-year time frame, based on our analysis. Scenario analysis by our analysts indicates that the market is pricing in an excessively optimistic outlook, limiting upside potential and leaving the investor exposed to Capital loss.

Other Definitions

Last Price: Price of the stock as of the close of the market of the last trading day before date of the report.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments,

Research Methodology for Valuing Companies

and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

Capital Allocation (or Stewardship) analysis published prior to Dec. 9, 2020, was determined using a different process. Beyond investment strategy, financial leverage, and dividend and share buyback policies, analysts also considered execution, compensation, related party transactions, and accounting practices in the rating.

Capital Allocation Rating: Our Capital Allocation (or Stewardship) Rating represents our assessment of the quality of management's capital allocation, with particular emphasis on the firm's balance sheet, investments, and shareholder distributions. Analysts consider companies' investment strategy and valuation, balance sheet management, and dividend and share buyback policies. Corporate governance factors are only considered if they are likely to materially impact shareholder value, though either the balance sheet, investment, or shareholder distributions. Analysts assign one of three ratings: "Exemplary", "Standard", or "Poor". Analysts judge Capital Allocation from an equity holder's perspective. Ratings are determined on a forward looking and absolute basis. The Standard rating is most common as most managers will exhibit neither exceptionally strong nor poor capital allocation.

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Sustainalytics ESG Risk Rating Assessment: The ESG Risk Rating Assessment is provided by Sustainalytics; a Morningstar company.

Sustainalytics' ESG Risk Ratings measure the degree to which company's economic value at risk is driven by environment, social and governance (ESG) factors.

Sustainalytics analyzes over 1,300 data points to assess a company's exposure to and management of ESG risks. In other words, ESG Risk Ratings measures a company's unmanaged ESG Risks represented as a quantitative score.

Unmanaged Risk is measured on an open-ended scale starting at zero (no risk) with lower scores representing less unmanaged risk and, for 95% of cases, the unmanaged ESG Risk score is below 50.

Based on their quantitative scores, companies are grouped into one of five Risk Categories (negligible, low, medium, high, severe). These risk categories are absolute, meaning that a 'high risk' assessment reflects a comparable degree of unmanaged ESG risk across all subindustries covered.

The ESG Risk Rating Assessment is a visual representation of Sustainalytics ESG Risk Categories on a 1 to 5 scale. Companies with Negligible Risk = 5 Globes, Low Risk = 4, Medium Risk = 3 Globes, High Risk = 2 Globes, Severe Risk = 1 Globe. For more information, please visit sustainalytics.com/esg-ratings/

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